Glivenko – Cantelli Theorem:

$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \xrightarrow[n \to \infty]{a.s.} 0$. [max error is going to 0, as $n \to \infty$]

Application II: $X_1, X_2, \dots$ iid $\sim L^2(\mathbb{P})$

sample mean: $\bar{X}_n = \frac{X_1 + \cdots + X_n}{n} \xrightarrow[n \to \infty]{\mathbb{P}} \mathbb{E}X_1$

$(\bar{X}_n)^2 \xrightarrow[n \to \infty]{\mathbb{P}} (\mathbb{E}X_1)^2$ by continuous mapping theorem

sample variance: $S^2_n = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2$, $n \geq 2$.

S_n: sample standard deviation.

check: $S_n^2 = \frac{n}{n-1} \cdot \frac{1}{n} \sum_{j=1}^{n} X_j^2 - \frac{n}{n-1} (\bar{X}_n)^2$

$\frac{1}{n} \sum_{j=1}^{n} X_j^2 \xrightarrow{\mathbb{P}} \mathbb{E}X_1^2$.

$(\bar{X}_n)^2 \xrightarrow{\mathbb{P}} (\mathbb{E}X_1)^2$.

$E(X_j^2) < \infty$ as $X_1, X_2, \dots$ are in $\mathbb{E}X_1^2$. $L^2(\mathbb{P})$.

$X_1^2, X_2^2, \dots$ iid with finite mean.

$S_n^2 \xrightarrow[n \to \infty]{\mathbb{P}} \mathbb{E}X_1^2 - (\mathbb{E}X_1)^2 = \operatorname{var} X_1$

$S_n = \sqrt{S_n^2} \xrightarrow[n \to \infty]{\mathbb{P}} \sqrt{\operatorname{var} X_1}$: by continuous mapping theorem.

$f: \mathbb{R} \to \mathbb{R}$, $f(x) = \begin{cases} \sqrt{x} & \text{if } x \geq 0 \\ 0 & \text{o.w.} \end{cases}$

Recall: $X \sim \mu$ (i.e. X is an RV with law μ).

$F_X(x) = \mu((-\infty, x])$ $x \in \mathbb{R}$

$F_X(x) = \alpha F_c(x) + \beta F_d(x)$. ——— (*) $\alpha, \beta \geq 0$, $\alpha + \beta = 1$.

F_c: continuous DF; F_d: purely continuous or Discrete DF.

Fact: F_c is differentiable (λ) almost everywhere.

we can rewrite (*) as $\mu = \alpha \mu_c + \beta \mu_d$

(μ_c: continuous) (μ_d: discrete)

probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Focus on F_c or μ_c.

Application of Lebesgue decomposition theorem:

$\mu_c = \gamma \mu_c' + \delta \mu_c^2$ with $\gamma, \delta \geq 0$, $\gamma + \delta = 1$.

μ_c', μ_c^2: probability measures.

s.t: $\mu_c' \perp \lambda$ (singularity) and $\mu_c^2 \ll \lambda$ (absolute continuity)

$\exists A \in \mathcal{B}_{\mathbb{R}}$ s.t. $\mu_c'(A) = 0$, $\lambda(A) > 0$.

$\mathbb{R} = A \cup A^c$ and $\mu_c'(A^c) > 0$, $\lambda(A^c) = 0$? singular behaviour.

$\forall A \in \mathcal{B}_{\mathbb{R}}$ with $\lambda(A) = 0$ then $\mu_c^2(A) = 0$.

$\mu = \alpha \mu_c + \beta \mu_d = \alpha(\gamma \mu_c' + \delta \mu_c^2) + \beta \mu_d$ → convex combination.

$= \alpha \gamma \mu_c' + \alpha \delta \mu_c^2 + \beta \mu_d$

$\alpha \gamma \mu_c'$: singular continuous; $\alpha \delta \mu_c^2$: absolutely continuous; $\beta \mu_d$: discrete.

$F_x(x) = \alpha \gamma F_c'(x) + \alpha \delta F_c^2(x) + \beta F_d(x)$

$F_c'(x)$: singular continuous cdf; $F_c^2(x)$: abs continuous

↳ $F'(x) = 0$ $\lambda(a.e)\, x$.

Devil's staircase.

Radon-Nikodym Theorem:

If $\mu \ll \nu$ // σ finite measures.

Then $\mu(A) = \int_A f \, d\nu \quad \forall A \in \mathcal{F}$ for some $f: \Omega \to [0, \infty)$ measurable.

with $\int_\Omega f \, d\nu = \mu(\Omega) = 1$.

Radon Nikodym Thm:

If $\mu_1 \ll \mu_2$ (σ-finite), then $\exists f: \Omega \to [0,\infty)$ measurable such that $\mu_1(A) = \int_A f \, d\mu_2 \quad \forall A \in \mathcal{F}$ ——— (*)

From (*) $\boxed{\frac{d\mu_1}{d\mu_2} = f}$

for Radon Nikodym derivative/density.

$P \circ X^{-1} = \mu$

$= d(\underbrace{\alpha \mu_{ac}}_{\ll \lambda} + \delta \mu_{sc}) + \beta \mu_d.$

if (*) holds, $\int_{\mathbb{R}} g \, d\mu_1 = \int_{\mathbb{R}} g f \, d\mu_2$.

Examples of μ_{ac}

(i) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad x \in \mathbb{R}.$

$\mu_{ac}(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \, d\lambda(x)$

(ii) $f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & \text{ow} \end{cases}$

$\int_{\mathbb{R}} f(x) \, dx = 1$

$\mu_{ac}(A) = \int_A f(x) \, d\lambda(x).$

$X \sim N(0,1)$

$P \circ X^{-1}(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \, d\lambda(x)$

$\frac{d P \circ X^{-1}}{d\lambda}(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \forall x$

$\mathbb{E}\, g(X)$

$= \int_{\Omega} g(X(\omega))\, d\mathbb{P}(\omega)$

$= \int_{\mathbb{R}^2} g(x)\, d\mathbb{P}_X^{-1}(\omega)$

$= \int_{\mathbb{R}^2} g(x) \frac{1}{\sqrt{2\pi}} e^{-a^2/2}\, d\lambda(x)$

More generally;

$X \sim \mu$ (abs. continuous)

$d\mu = f\, d\lambda$

$\mathbb{E}X = \int_{\mathbb{R}^2} x\, d\mu(x)$

$= \int_{\mathbb{R}^2} x\, f(x)\, d\lambda(x).$

* Characteristic functions.

X: RV

$\Phi_X : \mathbb{R} \longrightarrow \mathbb{C}$

$\Phi_X(u) = \mathbb{E}\, e^{iuX}$

$= \mathbb{E}\cos(uX) + i\, \mathbb{E}\sin(uX) \quad \forall u \in \mathbb{R}.$

$-1 \le \cos(t) \le 1$

$-1 \le \sin t \le 1 \quad \forall t \longrightarrow \Phi_X$ is a bounded function

$|\Phi_X(u)| \le 1, \quad \forall u \in \mathbb{R}.$

If $u_n \xrightarrow[n\to\infty]{} u.$

$\cos(u_n X) \xrightarrow[n\to\infty]{a.s.} \cos(uX)$

If $u_n \longrightarrow u$ then $\cos(u_n X) \xrightarrow[n\to\infty]{a.s.} \cos(uX)$

$\sin(u_n X) \xrightarrow[n\to\infty]{a.s.} \sin(uX)$

as $-1 \le \cos t \le 1$

$-1 \le \sin t \le 1$ then by DCT:

$\mathbb{E}\cos(u_n X) \xrightarrow{n\to\infty} \mathbb{E}\cos(uX)$

$\mathbb{E}\sin(u_n X) \xrightarrow{n\to\infty} \mathbb{E}\sin(uX)$

$\lim_{n\to\infty} \Phi_x(u_n) = \Phi_x(u)$

$\lim_{n\to\infty} \Phi_x(u_n) = \Phi_x(u)$: sequential criterion of continuity

So $\Phi_x : \mathbb{R}^k \to k$ is continuous.

Ex : $\Phi_x : \mathbb{R} \to \mathbb{C}$ is uniformly continuous.

In Higher dimension

For Random vectors $X = (X_1, \dots, X_p)^t$

$\Phi_X : \mathbb{R}^p \to \mathbb{C}$

$\Phi_X(u) = \mathbb{E}\, e^{iuX} \quad \forall\, u \in \mathbb{R}^p$.

X, Y - R.V.s if X and Y are independent:

then $\Phi_{X,Y}(u,v) = \Phi_X(u)\,\Phi_Y(v) \quad \forall u, v \in \mathbb{R}$.

↳ can be extended to n dimension.

Proposition: Suppose $\mathbb{E}|X| < \infty$ equivalently $X \in L^1(\mathbb{P})$

then Φ_X is continuously differentiable with

$\Phi_X'(u) = \int ix\, e^{iux}\, d\mathbb{P}\circ X^{-1}(x) = i\,\mathbb{E}(X e^{iuX})$

Problem set 12

Q4

Given $\begin{cases} X_n \xrightarrow[n\to\infty]{L^1(\mathbb{P})} X \\ Y_n \xrightarrow[n\to\infty]{L^\infty(\mathbb{P})} Y \end{cases}$

Show $X_n Y_n \xrightarrow[n\to\infty]{L^1(\mathbb{P})} XY$

$|Y| \le^{a.s.} \|Y\|_\infty$

$\|XY\|_1 = \mathbb{E}|XY| \le \|Y\|_\infty \mathbb{E}|X| = \|Y\|_\infty \|X\|_1$

$\frac{1}{p} + \frac{1}{q} = 1 \longrightarrow \|XY\|_1 \le \|X\|_p \|Y\|_q$.

Problem set 8

Given: i) $X: (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ R.V.

ii) $Y: \Omega \longrightarrow \mathbb{R}$ Function

iii) $X = Y$ a.s.

$\exists A \in \mathcal{F}, \mathbb{P}(A) = 0, \quad \forall \omega \in A^c, \; X(\omega) = Y(\omega)$.

$y \in \mathbb{R}$.

$$(Y \le y) = ((Y \le y) \cap A^c) \cup ((Y \le y) \cap A)$$

$$= ((X \le y) \cap A^c) \cup \underbrace{((Y \le y) \cap A)}_{\text{Subset of } A.}$$

Subset of null set.

*completion of a probability space:

Given $(\Omega, \mathcal{F}, \mathbb{P})$

Look at all sets $A \in \mathcal{F}$ with $\mathbb{P}(A) = 0$

Look at all subsets $B \subseteq A$. Is $B \in \mathcal{F}$?

Example: $\Omega = \{1, 2, 3\}$

$\mathcal{F} = \langle \phi, \{1\}, \{2, 3\}, \Omega \}$.

$\delta_1(\{2, 3\}) = 0$.

$A = \{2, 3\}, \quad B = \{2\} \notin \mathcal{F}$.

Derivatives of characteristic function:

If $X \in L^1(\mathbb{P})$ then $\Phi'_X(u) = i\,\mathbb{E}[X e^{iuX}]$

More generally, $X \in L^k(\mathbb{P})$ (i.e. $\mathbb{E}|X|^k < \infty$) the

$$\Phi_X^{(n)}(u) = i^n \mathbb{E}(X^n e^{iuX}) \quad 1 \le n \le k$$

$$\Phi_X^{(n)}(0) = i^n \mathbb{E}X^n$$

$$\Phi_X(u) = \mathbb{E}(e^{iuX})$$

C^k: derivatives upto k.

k^{th} order derivative continuous.

Thm: Suppose $X \in \mathcal{L}^k(\mathbb{P})$. k — positive integer, then:

$$\Phi_X(u) = \sum_{j=0}^{k} \frac{i^j}{j!} (\mathbb{E}X^j) u^j + o(|u|^k) \text{ as } u \to 0.$$

i.e. $$\lim_{u \to 0} \frac{\Phi_X(u) - \sum_{j=0}^{k} \frac{i^j}{j!} (\mathbb{E}X^j) u^j}{|u|^k} = 0.$$

Thm: ~~Formula~~ Inversion formula for characteristic functions.

$\mathcal{L}aw$ — probability measure on $\mathbb{R}$.

$F_X : DF$, $F_X(x) = \mathcal{L}((-\infty, x])$ $x \in \mathbb{R}$.

Then $F_X(b) - F_X(a) = \mathcal{L}([a,b])$ $= \lim_{c \to \infty} \frac{1}{2\pi} \int_{-c}^{c} \frac{e^{-iua} - e^{-iub}}{iu} \Phi_X(u) du$ ~~$\forall a < b$~~ for all $a < b$, where F_X continuous.

Thm: The characteristic fn uniquely determines the law of an RV. i.e. if $\Phi_X = \Phi_Y$ then $X \overset{d}{=} Y$.

$$\overline{\Phi_X(u)} = \mathbb{E}\left(\overline{e^{iuX}}\right) = \mathbb{E}(e^{-iuX}) = \Phi_{-X}(u) = \Phi_X(u)$$

Given X: Find $Y \overset{d}{=} -X$, Y independent of X.

$$\Phi_{X+Y}(u) = \Phi_X(u)\Phi_Y(u) = |\Phi_X(u)|^2$$

$$= \Phi_X(u)\Phi_{-X}(u) = \overline{\Phi_X(u)}\Phi_X(u).$$